

Name: _____

Period: _____

Date: _____

Algebra II H

Unit 7 - Rational Functions - Study Guide**Assessment Objectives:**

- ★ I can rewrite rational functions using polynomial division.
- ★ I can graph a rational function by identifying the vertical asymptote, horizontal asymptote, x-intercept, y-intercept, domain, range.
- ★ Given a graph of a rational function, I can identify the vertical asymptote, horizontal asymptote, x-intercept, y-intercept, domain, range.
- ★ I can write and use rational functions that represent average cost.
- ★ I can simplify, multiply, and divide rational expressions.
- ★ I can add and subtract rational expressions.
- ★ I can simplify complex rational expressions.
- ★ I can solve a rational equation by
 - Cross multiplication
 - Multiplying by the least common denominator
- ★ I can explain how extraneous solutions may arise when solving a rational equation.
- ★ I can identify extraneous solutions when solving a rational equation.

Prior Learning Skills:

- ★ I can factor using: Greatest common factor (GCF), Factoring when $a=1$, Factoring by grouping when $a \neq 1$, Difference of Two Squares, Difference of Cubes, Sum of Cubes

Identify horizontal and vertical asymptote, domain, and range, and x- and y-intercepts. Then, sketch the graph.

1. $f(x) = \frac{3}{x}$

3. $g(x) = \frac{4}{x} + 3$

2. $g(x) = \frac{-12}{x}$

4. $y = \frac{3}{x-4} - 1$

Rewrite $g(x)$ in the form $g(x) = \frac{a}{x-h} + k$ where h and k are constants. Then identify asymptotes, domain and range, and x- and y-intercept. Then, sketch a graph.

5. $g(x) = \frac{4x+3}{x+1}$

7. $f(x) = \frac{2x-6}{x+2}$

6. $h(x) = \frac{4x-1}{x+1}$

8. $y = \frac{2x+1}{x-3}$

If possible, simplify the rational expression.

9. $\frac{3x^2-5x-2}{x^2-4}$

10. $\frac{x^2-25}{x^3-125}$

Multiply the rational expressions. Simplify the result.

11. $\frac{3x^2-12}{5x-10} \cdot \frac{1}{2x+4}$

12. $\frac{21x^{10}y^5}{5x^2} \cdot \frac{x^3}{35y^4}$

Divide the rational expressions. Simplify the result.

13. $(x^2 + 10x - 24) \div \frac{x^2 - 144}{3x - 36}$

14. $\frac{x^3 - 8}{64x} \div \frac{x^2 - x - 2}{16x^2}$

Simplify fully.

15. $\frac{\frac{x+6}{5} + \frac{1}{x}}{\frac{5}{x} + \frac{25}{x^2}}$

17. $\frac{\frac{1}{3x^2 - 3}}{\frac{5}{x+1} - \frac{x+4}{x^2 - 3x - 4}}$

16. $\frac{\frac{1}{2x-5} - \frac{7}{8x-20}}{\frac{x}{2x-5}}$

Solve the equation. Check for extraneous solutions.

18. $\frac{x}{x^2 - 10} = \frac{3}{2x + 1}$

22. $\frac{x}{5x+5} = \frac{1}{x+2} + \frac{1}{x^2 + 3x + 2}$

19. $\frac{3}{x} - \frac{2}{x+1} = \frac{4}{x}$

23. $\frac{3}{x^2} = \frac{x-4}{3x^2} + \frac{2}{3x^2}$

20. $\frac{2}{2x+5} + \frac{3}{2x-5} = \frac{5x+5}{4x^2 - 25}$

24. $\frac{1}{x} + \frac{3x+12}{x^2 - 5x} = \frac{7x-56}{x^2 - 5x}$

21. $\frac{1}{x-2} + \frac{1}{x+3} = \frac{5}{x^2 + x - 6}$

25. $\frac{2}{x^2 - 2x - 3} = \frac{3x}{x+1} + \frac{1}{2x-6}$

- 26.** You are creating statues made of cement. The mold costs \$300. The materials for each statue costs \$22.
- Estimate how many statues must be made for the average cost per statue to fall below \$30.
 - What happens to the average cost as more statues are created?
- 27.** The senior class is sponsoring a dinner. The cost of catering the dinner is \$9.95 per person plus an \$18 delivery charge.
- Write a model that gives the average cost per person.
 - Estimate the number of people needed to lower the cost to \$11 per person.
 - Find and interpret the horizontal asymptote of your model.
 - Find and interpret the vertical asymptote of your model.
- 28.** The average cost, in dollars, per unit $C(x)$ to produce x units of plywood is given by $C(x) = \frac{300}{x+10}$.
- What is the cost per unit when 590 units are produced?
 - If the cost per unit is \$1.50, how many units have been produced?
- 29.** A rare species of insect was discovered in the rainforest of Costa Rica. Environmentalists transplant the insect into a protected area. The population of the insect t months after being transplanted is $P(t) = \frac{45(1+0.6t)}{(3+0.02t)}$.
- What was the population when $t = 0$?
 - What will the population be after 10 years?
 - When will there be 549 insects?
- 30.** For a fundraising project, your math club is publishing a graphic art calendar. The cost of the digital images and the permission to use them is \$850. In addition to these "one-time" charges, the unit cost of printing each calendar is \$3.25.
- Write a model that gives the average cost per calendar as a function of the number of calendars printed.
 - Estimate the number of calendars you need to print before the average cost drops to \$5 per calendar.
 - Describe what happens to the average cost as the number of calendars printed increases.
 - Suppose you decide to generate your own designs on a computer to save money. The cost for the software (a "one-time" cost) is \$125. Write a model that gives the average cost per calendar as a function of the number of calendars printed. Estimate the number of calendars you need to print before the average cost drops to \$5 per calendar.